Core Finance
Week 9

MGMP 543

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Spring 2025

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Welcome Back! Today's Topics

- Boeing 7E7 Case
- Week 8 Recap
- Quiz 7
- Week 7 Examples
- Discounted Cash Flow (DCF)
  - Free Cash Flow to the Firm (FCFF)
- Payout Policy
- Examples

Boeing 7E7 Case

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Case Questions

1. What is an appropriate required rate of return against which to evaluate the prospective IRRs from the Boeing 7E7?
   a) Which beta did you use? Why?
   b) When you used the CAPM, which risk premium and risk-free rate did you use? Why?
   c) Which capital structure weights did you use? Why?
2. Judged against your WACC, how attractive is the Boeing 7E7 project?
   a) Under what circumstances is this project economically attractive?
   b) What does a sensitivity analysis (either the analysis presented in the case or another that you have done on your own) reveal about the nature of Boeing's gamble on the 7E7?
   c) Should Boeing have launched the 7E7 in 2003?

Week 8 Recap

What Did We Learn Last Week?

Quiz 7

Q2. If firms did not have limited liability, their asset risk would be increased. (Hint: Assume perfect capital markets). True or False?

Q3. As long as the firm is certain that the return on assets will be higher than the interest rate, an issue of debt makes the shareholders better off. True or False?

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Q4. In perfect capital markets, according to Modigliani and Miller Proposition II, the firm's expected return on assets depends on several factors including the firm's capital structure. True or False?

Q5. Modigliani and Miller Proposition II states that the rate of return required by shareholders increases steadily as the firm's debt-equity ratio increases. True or False?

Week 7 Examples Continued (See Last Week's Slides)

Discounted Cash Flow (DCF)

Aim: Value any project/firm

- Weeks 1-5: Tools for valuing assets
- Week 6: Hopefully, prices are right 😅
- Week 8: Compare financing options
- **Week 9: Putting this together (with some accounting) to value any project/firm!**

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Free Cash Flow to the Firm (FCFF)

Valuing Dividend-Paying Firms (Recap)

Week 3 introduced Dividend Discount Model (DDM):

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

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Weeks 5 & 8 learnt how to calculate return on equity:

$$E[r_E] = r_{rf} + \beta_E\left(E[r_M] - r_{rf}\right)$$

We can use this return on equity as our discount rate in the DDM:

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r_E)^t}$$

What May Go Wrong Using DDM to Value Firms?

What May Go Wrong Using DDM to Value Firms?

- **Dividends are discretionary. Some firms may choose to pay no/low dividends. Are they worthless / less valuable firms?**
- **PVGO taught us that firms can increase their growth by reinvesting cash in NPV>0 projects rather than paying out dividends. Are firms that pay less in dividends worth less?**

There must be a better way to value firms…

Discounted Cash Flow (DCF)

Now we have a general formula to value all types of firms irrespective of whether they pay dividends or how they are financed:

$$Firm\ Value = \sum_{t=1}^{\infty} \frac{FCFF_t}{(1 + WACC)^t}$$

Discounted Cash Flow (DCF)

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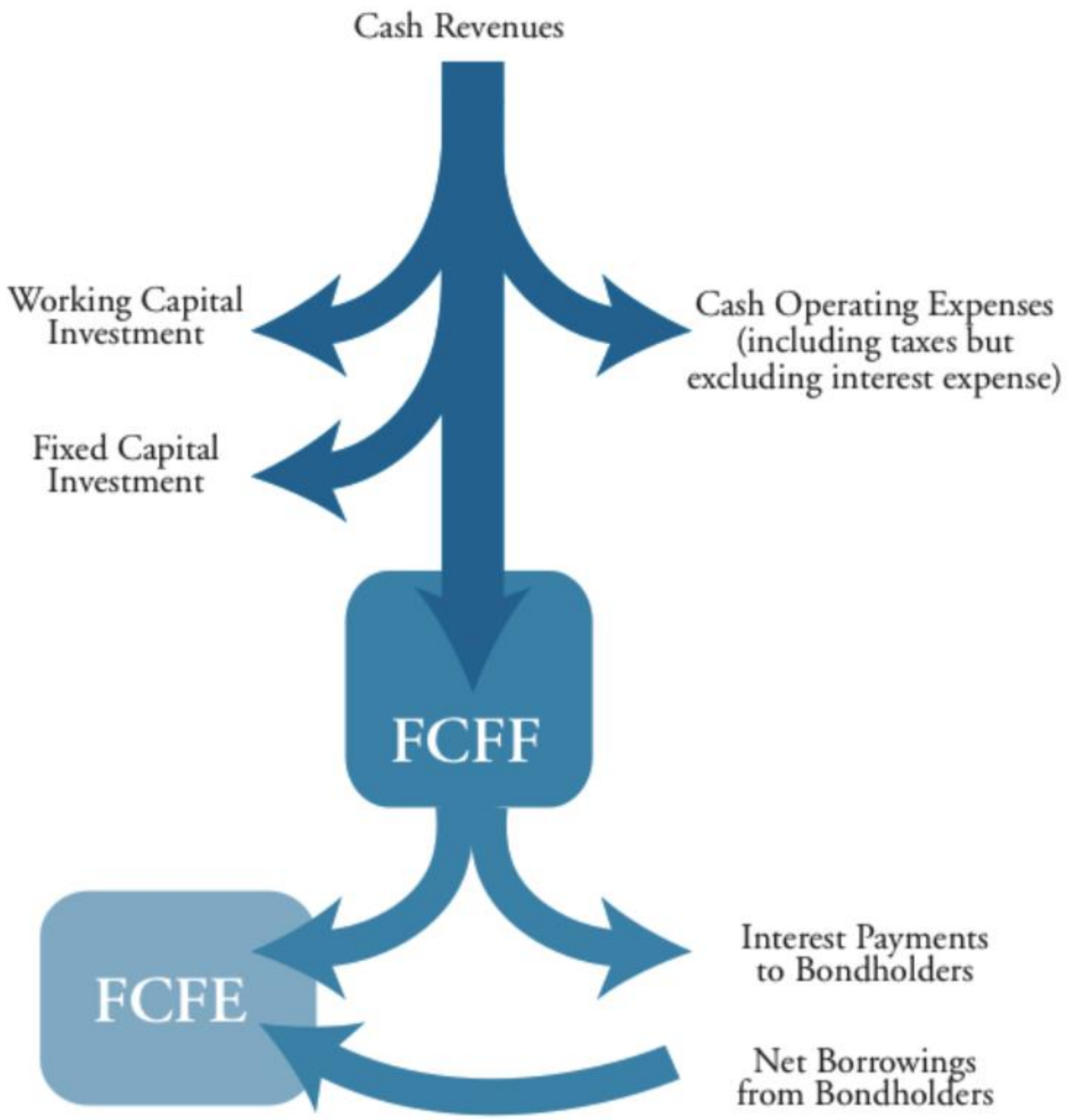
$$Firm\ Value = \sum_{t=1}^{\infty} \frac{FCFF_t}{(1 + WACC)^t}$$

- We focus on real economic cash flows ***NOT*** accounting profit
- $FCFF$ **estimates the cash generating ability of a firm**
- **Our discount rate is the opportunity cost to the firm (e.g., WACC if the project has a similar risk to the firm's risk)**

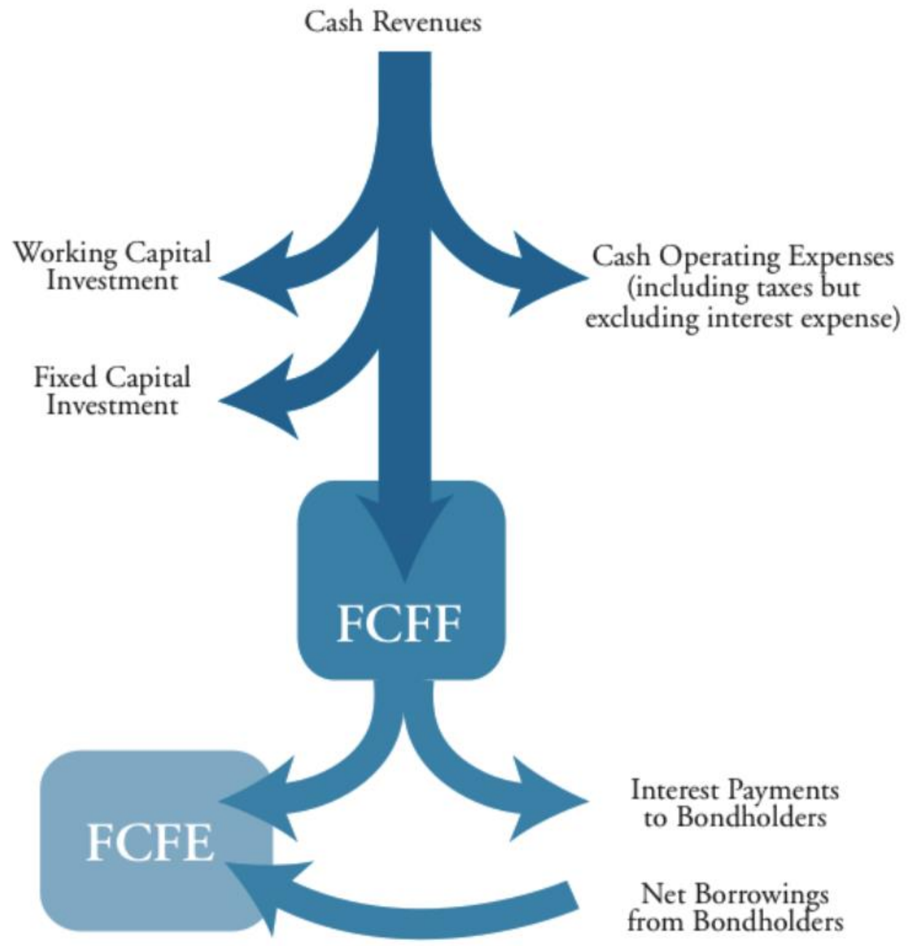
Free Cash Flow to the Firm (FCFF)

Free Cash Flow to the Firm (FCFF) captures all cashflows generated from operations available to all investors of the firm

Free Cash Flow to Equity (FCFE) cash available to distribute to shareholders.



Calculating
Free Cash Flow to the Firm (FCFF)



$$\boldsymbol{FCFF} = [EBIT \ \times (1 - tax\ rate)] + Dep - FCInv - WCInv$$

- EBIT: Earnings Before Interest and Taxes (EBIT also known as “Operating Profit”)
- Dep: Depreciation (n.b. assuming this is the only ‘Non-Cash Charge’)
- FCInv: Fixed Capital Investment
- WCInv: Working Capital Investment

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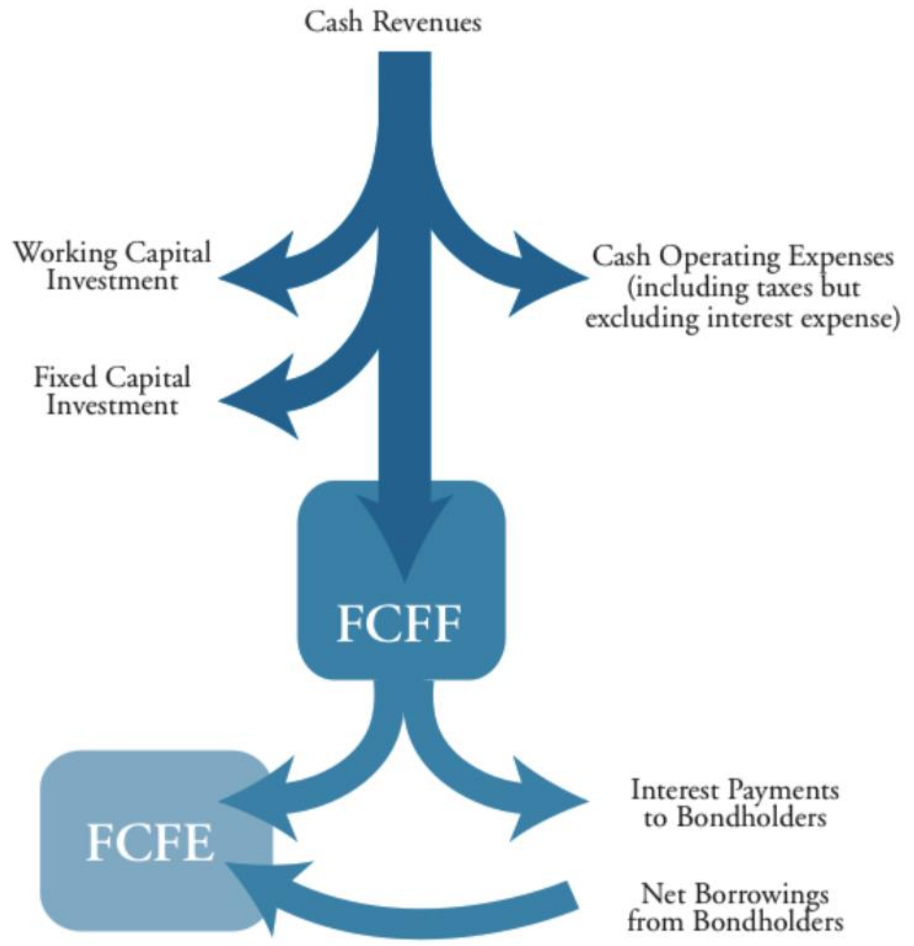

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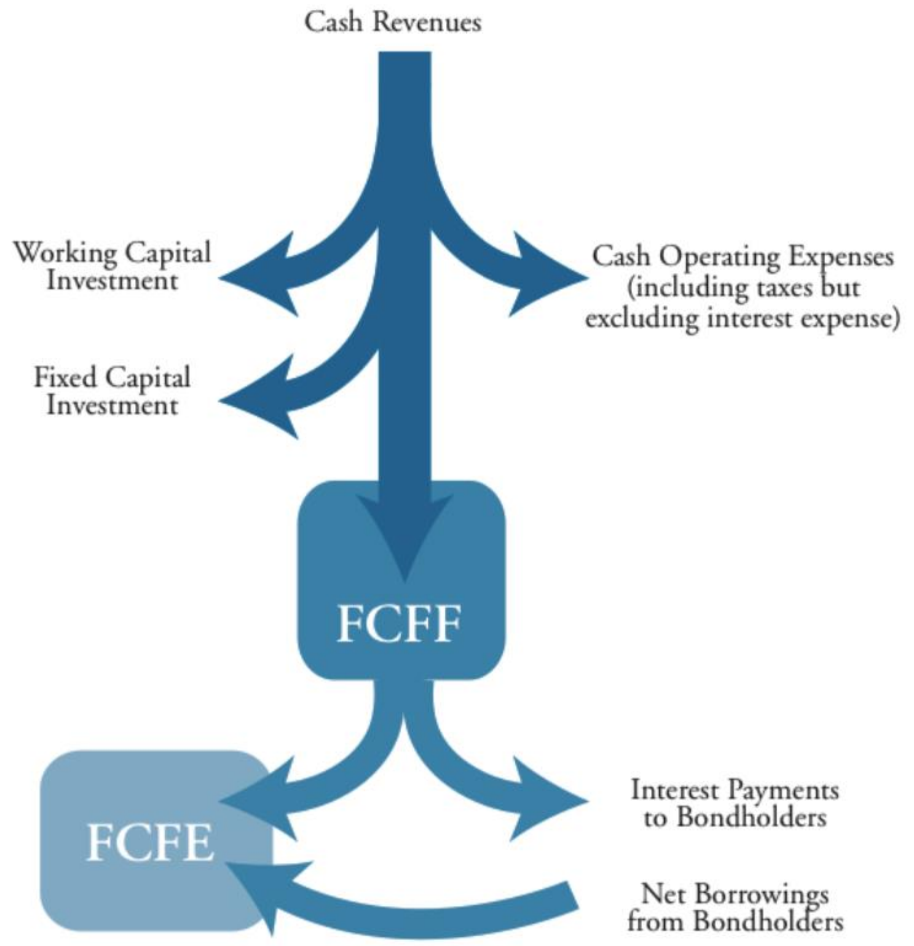
There are other ways to calculate FCFF that you don’t need to know for this class:

$$FCFF = NI + NCC + [Int \times (1 - tax\ rate)] - FCInv - WCInv$$
$$FCFF = [EBIT \times (1 - tax\ rate)] + Dep - FCInv - WCInv$$
$$FCFF = [EBITA \times (1 - tax\ rate)] + (Dep \times tax\ rate) - FCInv - WCInv$$
$$FCFF = CFO + [Int \times (1 - tax\ rate)] - FCInv$$

Calculating **EBIT** and **Dep in** Free Cash Flow to the Firm (FCFF)



$$\boldsymbol{FCFF} = [\boldsymbol{EBIT} \times (\mathbf{1} - \boldsymbol{tax\,rate})] + \boldsymbol{Dep} - FCInv - WCInv$$

- **EBIT** is pre-tax income after paying for Cost of Goods Sold (COGS) , Selling, General, and Administrative Expenses (SGA, i.e., overheads and operating costs), and accounting for deprecation (& other 'Non-Cash Charges' e.g., amortization of intangible assets, asset impairments, bad debt provisions).

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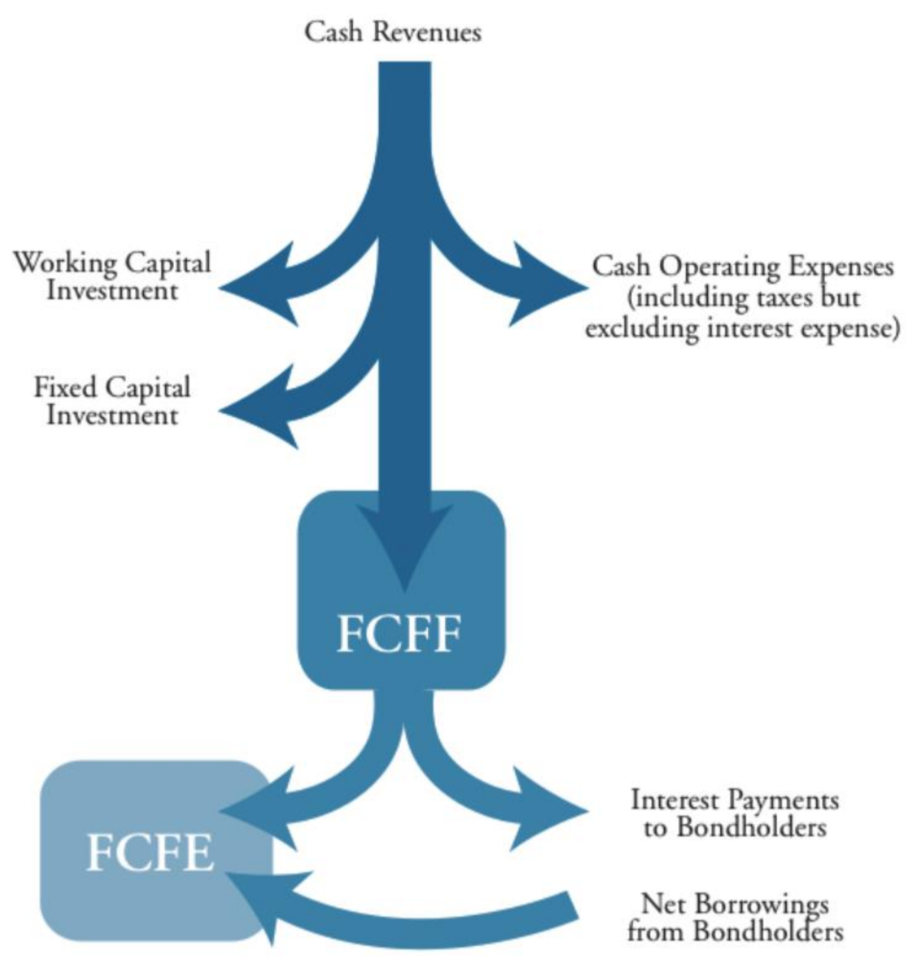

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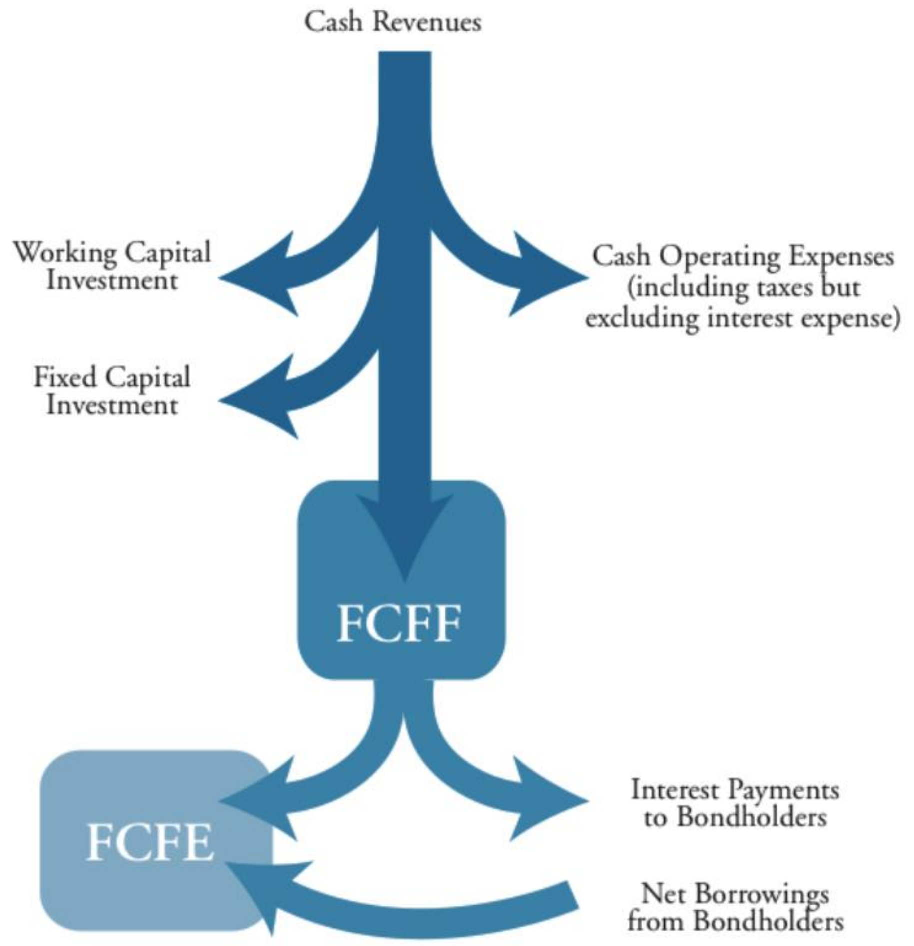
- Need to add back in **depreciation** (and other non-cash charges) as no cash has left the company – this is just an accounting adjustment.

Calculating **FCInv** in Free Cash Flow to the Firm (FCFF)

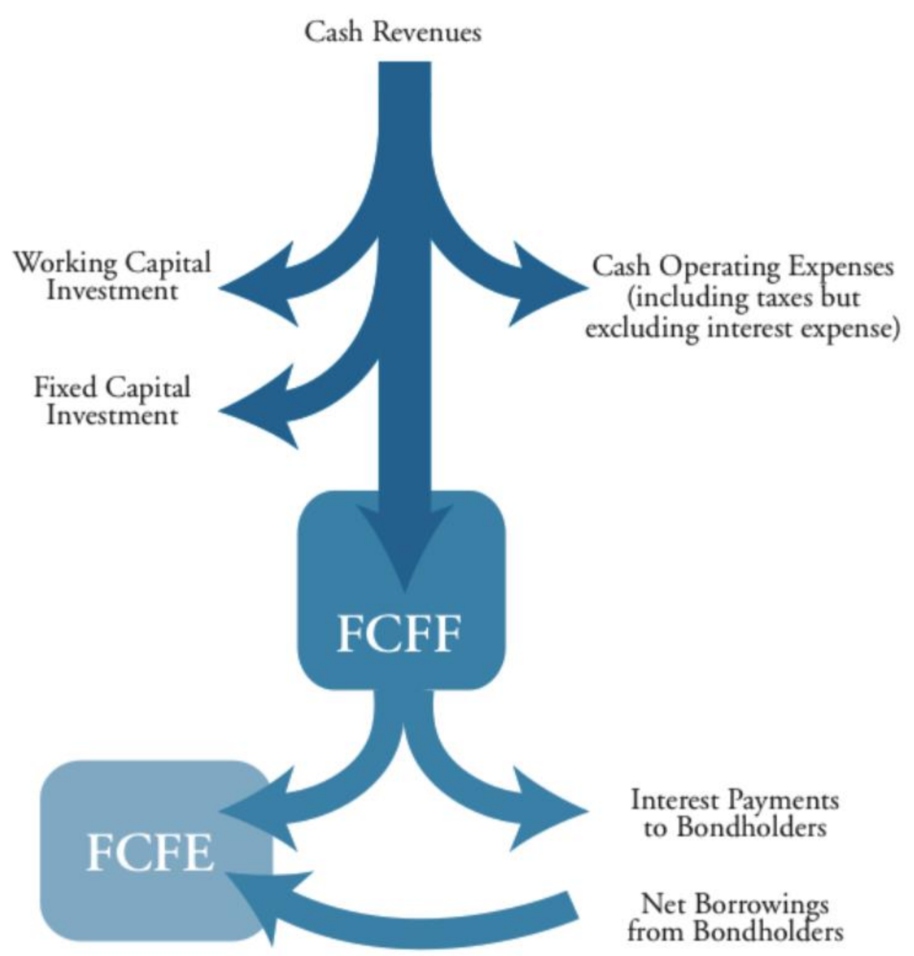


$$FCFF = [EBIT \times (1 - tax\ rate)] + Dep - FCInv - WCInv$$

- Fixed Capital Investment (FCInv) doesn’t appear on Income Statement, but represents cash leaving a firm
- If no fixed assets were sold, can calculate using difference in property, plant, and equipment (PP&E):

$$FCInv = Capital\ Expenditures = Ending\ Gross\ PP\&E - Beginning\ Gross\ PP\&E$$

Calculating **WCInv** in Free Cash Flow to the Firm (FCFF)

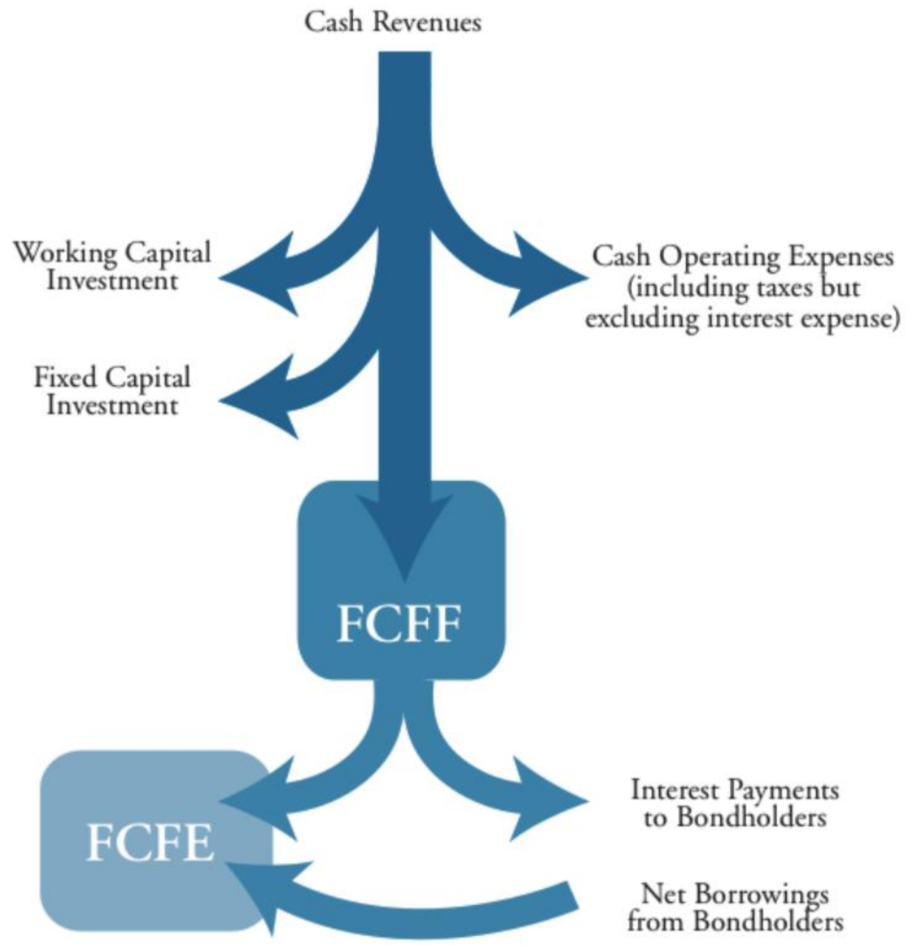


$$\boldsymbol{FCFF = [EBIT \times (1 - tax\ rate)] + Dep - FCInv - WCInv}$$

- Working Capital Investment (WCInv) is the change in working capital:

$$\boldsymbol{WCInv = WC_t - WC_{t-1}}$$

Calculating **WCInv** in Free Cash Flow to the Firm (FCFF)



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- WC = Working Current Assets (WCA) – Automatic Sources (AS)
- Working Current Assets (WCA) = Current Assets – Excess Cash = Necessary Cash + Accounts Receivable + Inventory + Other Current Assets
- Automatic Sources (AS) = Accounts Payable + Anything “Accrued” (e.g., expenses, taxes, wages). AS are non-interest bearing liabilities.

Discounted Cash Flow (DCF)

Enterprise Value (EV)

- **'Enterprise Value (EV)'** measures the total value of the firm if you were to buy all the financial claims of the firm. I.e., it is the value of the entire firm.

$$EV = \text{Market Capitalization} + \text{Market Value of Debt} - \text{Cash and Equivalents}$$

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  EV = Market Capitalization + Market Value of Debt – Cash and Equivalents

- And if other financing sources you include them as well.
- Cash & equivalents ***decrease*** EV because it is a rebate on the purchase price (e.g., you buy jeans for $100 and they have $20 cash in the pocket.

  then really the jeans cost $80)

- Can rearrange this to back out the value of equity:

  Market Capitalization = EV – Market Value of Debt + Cash and Equivalents

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Steps to Valuing a Firm Using Discounted Cash Flow (DCF)

$$Firm\ Value = \sum_{t=1}^{\infty} \frac{FCFF_t}{(1+WACC)^t}$$

1. Project the future free cash flows to the firm (FCFF) over the forecast period ‘H’ (e.g., 5 years typical)

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2. Calculate the weighted average cost of capital (WACC) to use to discount FCFF
3. Project a ‘**terminal value**’ – the present value of free cash flows after the forecast period (PV_H)
4. Discount the free cash flows to get a value of the firm
   - For value of equity, Enterprise Value – Firm Value
   - For share price, $\frac{\text{Value of Equity}}{\text{Number of Shares Outstanding}}$

$$PV = \sum_{t=1}^{H} \frac{FCFF_t}{(1 + WACC)^t} + \frac{PV_H}{(1 + WACC)^H}$$

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Payout Policy

Measuring Dividend Policy

Dividend Payout = Dividends Paid / Earnings

- Fraction of earnings paid out to shareholders

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Dividend Yield = Dividends Paid / Stock Price

- Return an investor makes from dividends alone (i.e., not capital gains)
- Can think of this like an interest rate on debt, except dividend policy can change

Does Dividend Policy Matter for Stock Price or Capital Structure?

In Modigliani & Miller Theory, no!

- Higher returns are what investors care about.
- Investors should not care whether they get that in dividends or from stock price appreciation
  This is in theory, where there are not market imperfections.

When & How Should a Firm Return Cash to Shareholders?

- Invest in all NPV > 0 projects that return more than the firm's hurdle rate
- Return leftover cash to shareholders. How to do so?
  1. **Dividends**
     - Regular Dividends: Regular payment usually from profits.
       - Regular dividends are sticky over time, and they follow earnings (i.e., ⬆️ earnings, expect ⬆️ dividends)
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     - Company distributes newly created shares to all shareholders (similar to stock splits)
     - Increases number of shares, reduces earnings per share (EPS)
     - Not a true dividend because not paid in cash

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   - Not a true dividend because not paid in cash
3. **Share Repurchases / Stock Buybacks**
   - Company buys back its own shares directly from the market, or company offers shareholders they option to sell their shares as a fixed price
   - Reduces number of shares, increases earnings per share (EPS). Shares held as 'Treasury shares'
   - Increases firm leverage

Share Repurchases

- Firms may want to repurchase shares if the stock price is less than its fair value
- If so, you can create value for existing shareholders
    - E.g., If fair value is $100 and you repurchase for $80 then value is for shareholders is:

      ($100 - $80) x Number of Shares Repurchased

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Benefits?

- Positive signal? Management believes stock is undervalued
- More flexible alternative to dividends
- More tax efficient for tax-paying investors
  - Taxes are paid on dividends when they are received
  - Taxes through price appreciation are delayed until investor sells

  (taxes in the future are less in present value terms than an equivalent tax amount today)

Costs?

- Often ill-timed. Companies more likely to buy back stocks when the company is in good financial health and the stock price is high. If overvalued, buybacks reduce value!
- As with dividends, could cash be better spent on NPV > 0?
- Negative signal? May indicates that there are not other profitable growth opportunities

Is the Price Right? "The Dividend Disconnect" (Hartzmark & Soloman, 2019)

- Investors should care about total returns (dividends + capital gains)
- Dividends are highly predictable in timing and amount
- Dividends result in price decreases
- Investors trade as if dividends and capital gains are disconnected attributes
  - Dividends as small, stable gains. Prices as possible large gains/losses
  - Dividends and capital gains spent differently – dividends often not reinvested
  - Price changes not returns are salient for investors deciding whether to buy/sell

Examples

Q1. Calculate the free cash flows to the firm.

Sting's Income Statement

Income Statement

	2010 Forecast	2009 Actual
Sales	$300	$250
Cost of goods sold	120	100
Gross profit	180	150
SG&A	35	30
Depreciation	50	40
EBIT	95	80
Interest expense	15	10
Pre-tax earnings	80	70
Taxes (at 30%)	24	21
Net income	$56	$49

Balance Sheet

	2010 Forecast	2009 Actual
Cash	$10	$5
Accounts receivable	30	15
Inventory	40	30
Current assets	$80	$50
Gross property, plant, and equipment	400	300
Accumulated depreciation	(190)	(140)
Total assets	$290	$210
Accounts payable	$20	$20
Short-term debt	20	10
Current liabilities	$40	$30
Long-term debt	114	100
Common stock	50	50
Retained earnings	86	30
Total liabilities and owners' equity	$290	$210

Cash Flow From Operations Forecast for 2010

Net income	$56
+ depreciation	50
– WCInv	25
Cash flow from operations	$81

Q2. Calculate the free cash flows to the firm in year 0 to starting a Chicago Hot Dog Business in Houston.
Assume no taxes, deprecation, or other operating expenses.

Financed with $50 of your own cash as equity
A $50 line of credit from Brewers Bank that you borrow $30 on
Cost of operations: $80 cart bought with cash on hand
No hot dogs made or sold yet

Q3. How would your answer to the previous question change if you now contributed $10 less in equity and borrowed $10 more on line of credit?

Q4. Calculate the free cash flows to the firm in **year 1** to Chicago Hot Dog Business in Houston.
Assume no taxes, deprecation, or other operating expenses.

Year 0:

- Financed with $50 of your own cash as equity
- A $50 line of credit from Brewers Bank that you borrow $30 on
- Cost of operations: $80 cart bought with cash on hand
- No hot dogs made or sold yet

Year 1:

- Spend $20 on supplies for 20 hot dogs
- Sell 10 hot dogs for $40 (half of your supplies)
- Keep remaining 10 hot dogs in inventory
- Borrow an additional $20 for supplies at the beginning of the year
- Use $20 in revenues to pay interest and principal on line of credit
- Pay yourself a dividend of $20

Q5. Calculate the free cash flows to the firm in **year 1** to Chicago Hot Dog Business in Houston.
Now assume depreciation of the cart by $5 during year 1.
Assume no taxes or other operating expenses.
Year 0:
- Financed with $50 of your own cash as equity
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Q6. Calculate the free cash flows to the firm in **year 2** to Chicago Hot Dog Business in Houston.
Assume no taxes, deprecation, or other operating expenses.

Year 0:

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- A $50 line of credit from Brewers Bank that you borrow $30 on
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- Keep remaining 10 hot dogs in inventory
- Borrow an additional $20 for supplies at the beginning of the year
- Use $20 in revenues to pay interest and principal on line of credit
- Pay yourself a dividend of $20

Year 2:

- **No hot dogs made.**
- **Sell last $10 of hot dogs for $10**
- **Sell cart for $80**
- **Use $90 in revenues to pay off remaining bank loan (principal and interest) and pay yourself a dividend with the remainder**

Q7. How much is the Chicago Hot Dog Business in Houston worth at the end of year zero (after you've bought the cart)? Suppose WACC is 15%

Q8. How much is the **equity worth in** Chicago Hot Dog Business in Houston worth at the end of year zero (after you've bough the cart)? Suppose WACC is 15%

Q9. Consider an investment in Benedict's Bricks

a) Market value of equity of $300 million and market value of debt of $120 million. Further, it holds $40 million of cash on its balance sheet. Further, it has 10 million shares outstanding. Assume the beta of the equity is 1.2 and the beta of the debt is 0.1. The market risk premium is 8% and the risk free rate is 3%. Assume that the Tax Rate is 30%. What is the WACC of the firm?

b) What is the enterprise value of the firm?

c) Suppose the FCFF estimates for the next five years are: $10, $12, $14, $16, $17, with the cashflow occurring one year from today. At the end of the five years, the FCFF is estimated to grow at a constant rate of 3%, inline with inflation. Given all this information, what is the value of the firm?

d) Given your answer to c), what is the equity value? Further, should the price per share be?

e) If the firm is currently trading at $18.50 per share, is it under/over/fairly priced?

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